

BS – 101 Engineering Physics

Lecture – 4

Engr. Safee Ullah



Point Charge Inside a Spherical Surface

- Gauss' Law provides a different way to express the relationship between electric flux and electric field.
- It was formulated by a 18th century mathematician Carl Friedrich Gauss (1777-1855).



Point Charge Inside a Spherical Surface

- Gauss's law states that the total electric flux through any closed surface (a surface enclosing a definite volume) is proportional to the total (net) electric charge inside the surface.

$$\Phi_E = EA = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} (4\pi R^2) = \frac{q}{\epsilon_0}$$

Point Charge Inside a Spherical Surface

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The same number of field lines and the same flux pass through both of these area elements.

